Inside the Spec

Real Scenarios. Working Artifacts.
Decisions That Matter.

One Status, Three Systems, Different Meanings

Cross-System Status Mapping
A return workflow with separate evidence for each result.

CASE 02 · STATUS MAPPING

“Completed” does not name the result.

A customer returns one item for USD 80.00. The store, warehouse, and payment provider can all report COMPLETED, while describing different objects.

Source / object	What COMPLETED establishes
OMS · return request	Accepted return processed; refund instruction registered.
WMS · inspection	Inspection finished. The result can be ACCEPTED or REJECTED.
PSP · refund	Provider confirms issuance. Bank credit is not independently confirmed.

> **Mapping each value to “Money refunded” loses the evidence behind it.**

CLOSED_NO_REFUND is OMS closure after WMS REJECTED. Store refusal before inspection is outside this case.

Worked illustrative specification, not a production incident. One item, one inspection, at most one full refund. USD 80.00 is example data. Operational sections document the dependencies of the mapping.

Keep facts under their owners.

Owned dimension	Allowed projected values
OMS · return_state	UNKNOWN; OPEN; CLOSED_REFUND_REGISTERED; CLOSED_NO_REFUND
WMS · inspection_state	UNKNOWN; PENDING; ACCEPTED; REJECTED
PSP · refund_state	UNKNOWN; PENDING; COMPLETED; FAILED

UNKNOWN means no valid evidence. Each source updates its own fact. The display function has four inputs: three facts and F, the unresolved reconciliation flag retained from history.

The raw inspection_result exists only for WMS COMPLETED. It must be ACCEPTED or REJECTED. IN_PROGRESS forbids that field; lowercase “accepted” is invalid.

> **REFUND_ISSUED requires warehouse acceptance, matching PSP completion, and no unresolved conflict.**

Case decision: late OMS OPEN does not veto issuance. Missing WMS acceptance does. An early PSP result is preserved while warehouse evidence is checked; a contradictory no-refund closure requires reconciliation.

Map the source value to its owned fact.

Source · raw status / result	Projected fact value
OMS · OPEN	OPEN
OMS · COMPLETED	CLOSED_REFUND_REGISTERED
OMS · CLOSED_NO_REFUND	CLOSED_NO_REFUND
WMS · IN_PROGRESS	PENDING
WMS · COMPLETED/ACCEPTED	ACCEPTED
WMS · COMPLETED/REJECTED	REJECTED
PSP · PENDING	PENDING
PSP · COMPLETED	COMPLETED
PSP · FAILED	FAILED

Generated from status-mapping.csv, the authoritative dictionary. UNKNOWN is absence of a valid snapshot, not a raw source value. WMS IN_PROGRESS forbids inspection_result.

Evaluate the first matching rule.

O = return_state; W = inspection_state; P = refund_state. P known means P != UNKNOWN. Nine ordered rules produce eight statuses.

Condition	Result
D01 · Unresolved reconciliation flag	NEEDS_RECONCILIATION
D02 · Rejection + refund evidence, or no-refund closure + acceptance/refund evidence	NEEDS_RECONCILIATION; latch flag
D03 · W = REJECTED	RETURN_REJECTED
D04 · W ∈ {UNKNOWN, PENDING} AND (P != UNKNOWN OR O ∈ {CLOSED_REFUND_REGISTERED, CLOSED_NO_REFUND})	AWAITING_CONFIRMATION
D05 · W = ACCEPTED; P = COMPLETED	REFUND_ISSUED
D06 · W = ACCEPTED; P = FAILED	REFUND_FAILED
D07 · W = ACCEPTED; P = PENDING	REFUND_PROCESSING
D08 · W = ACCEPTED; P = UNKNOWN	REFUND_NOT_CONFIRMED
D09 · Otherwise	RETURN_IN_PROGRESS

D02 precisely: (W=REJECTED AND (P known OR O=CLOSED_REFUND_REGISTERED)) OR (O=CLOSED_NO_REFUND AND (P known OR W=ACCEPTED)). D04 follows D02, so contradictions take precedence.

Bind identity before judging content.

Every snapshot carries source_system, object_type, object_id, return_id, event_id, revision, contract_version, occurred_at, and a full payload.

Validation boundary	Required response
A · Untrusted identity or linkage	Quarantine; integration incident. Do not change a return fact or flag.
B · Trusted identity, invalid content	Quarantine; preserve fact/baseline. Latch reconciliation unless the strictly stale-content exception applies.

PSP includes refund_id, amount_minor, and currency. refund_id must match the bound object. Amount and currency must exactly match the registered instruction. A wrong amount on the correct refund is B, not an ignored event.

The trusted registry exists before dispatch. Contract version is 1. Values are case-sensitive; required fields cannot be null. Unsupported versions on reliably bound objects are B.

Native provider identifiers may differ from internal IDs. The adapter verifies their binding. If the provider lacks monotonic full snapshots, the adapter must establish authoritative ordering; arrival order alone cannot define revisions.

A replay and an older revision differ.

Incoming snapshot	Decision
Same source + event_id, identical event	Return the recorded outcome; no duplicate action.
Same event_id, changed content	Integrity conflict; latch even if revision is old.
Valid revision below accepted baseline	Ignore as stale.
Equal revision, equal business payload	No-op even if event_id differs.
Equal revision, different payload	Keep fact; latch reconciliation.
Higher revision	Validate allowed transition; commit atomically.

Worked order, with WMS ACCEPTED: PSP PENDING rev 1 → COMPLETED rev 4 → first delivery of PENDING rev 3. The last event is stale; REFUND_ISSUED remains.

An identical replay of the rev 4 event is a deduplication no-op. Revision gaps are permitted because snapshots are complete. Revisions are comparable only within one source object.

> **Serialize updates per return or retry a projection-version conflict. One source must not erase another source’s fact.**

New ID + supported envelope + valid revision below baseline: malformed content creates an incident, without changing the projection. Missing ID, invalid revision, unknown contract, or event-ID collision cannot use this exception.

Issued is not an irreversible success.

PSP COMPLETED is nonterminal. A newer, valid FAILED snapshot is a normal late outcome, even days later. It replaces the provider fact without manual re-baseline.

PSP transition	Handling
PENDING → COMPLETED	Issuance confirmed; continue late-outcome monitoring.
PENDING → FAILED	Failure confirmed; open a payments case.
COMPLETED → FAILED	Accept late failure; create durable finance handoff.
COMPLETED → PENDING	Invalid transition; reconciliation.
FAILED → PENDING / COMPLETED	Invalid ordinary change; authoritative reconciliation required.
Same state, higher valid revision	Accept the new baseline, including FAILED → FAILED.

Complete OMS, WMS, PSP, and deduplication tables are in transition-and-deduplication.md. Terminal corrections require audited re-baseline.

A late failure is retained even when another conflict masks it in the UI. It still creates a payments case and finance handoff. Provider issuance, bank visibility, and ledger effects are separate facts.

Keep internal states out of the UI.

Internal enum	Customer label
RETURN_IN_PROGRESS	Return in progress
RETURN_REJECTED	Return not accepted
AWAITING_CONFIRMATION	Return status being checked
REFUND_NOT_CONFIRMED	Return status being checked
NEEDS_RECONCILIATION	Return status being checked
REFUND_PROCESSING	Refund processing
REFUND_FAILED	Refund needs attention
REFUND_ISSUED	Refund issued by provider

REFUND_ISSUED: “Your bank may not show the credit yet.” REFUND_FAILED: “We need to resolve an issue with your refund. Contact support with your return reference.”

Show supporting text with the label. Every view offers support with the return reference. REFUND_NOT_CONFIRMED also covers accepted items whose OMS instruction is not registered yet; its neutral copy makes no submission claim. Full RU/EN copy: display-mapping.md.

Operational boundary: dispatch.

Authoritative WMS ACCEPTED authorizes dispatch. OMS stores that evidence with intent, linkage, and outbox atomically. Before an unsent create, the worker checks that authorization is not revoked. An early PSP event can reflect delayed WMS delivery to the projection.

Dispatch state	Required behavior
READY → IN_FLIGHT	Acquire lease; persist attempt before create.
Provider response or linked webhook	Bind native refund ID; record ACKNOWLEDGED. This is not payment success.
Timeout or expired IN_FLIGHT lease	Move to UNCERTAIN; query the original operation.
Existing provider object found	Acknowledge it; do not create another refund.
Guarantee missing, expired, or retry limit reached	MANUAL_REVIEW; preserve uncertainty.

A temporary “not found” is not proof of non-execution. Create retries require identical parameters and a verified provider guarantee covering scope, retention, and concurrent calls.

Transient lookup failures retain UNCERTAIN and schedule backoff with jitter. Provider outages share an incident. Safe reads continue during manual review.

Transactional Outbox preserves intent; it does not establish exactly-once execution at PSP. Persist attempts, request hashes, provider references, lease tokens, retry limits, and dedupe_valid_until. Never rotate a key to escape an ambiguous outcome.

Correct the fact. Keep the obligation.

D02 denotes incompatible facts, not fraud. Rejection plus PSP FAILED still needs process reconciliation and finance assessment. FAILED does not prove no prior financial effects; assessment may conclude no posting is needed.

Re-baseline step	Constraint
Fetch authoritative source snapshots	Verify identity, full payload, amount, and currency.
Validate revision and correction	No lower baseline. Equal revision with changed data requires source-owner confirmation.
Replace fact and baseline atomically	Audit operator, reason, evidence, before/after values. Recheck concurrency.
Recompute all contradictions	Do not clear D02 with a comment alone.

Example: authoritative WMS REJECTED rev 3 can replace ACCEPTED rev 2. If PSP issuance still conflicts, the corrected warehouse fact is stored and reconciliation remains open.

> **A financially consequential correction also persists a finance-outbox record. ERP downtime must not block the truthful fact.**

Broker acknowledgment is not finance acceptance. Overdue delivery triggers escalation, not deletion. Parked records stay visible in backlog and aging metrics.

Closing reconciliation requires an accepted finance handoff where applicable and no remaining D02 conflict. Finance determines postings, compensation, or a no-posting decision. Re-baseline itself neither moves money nor creates a customer debt.

Use deadlines that match the stage.

Condition	Deadline and owner
Downstream evidence missing	D04: PSP/closed OMS implies final inspection. D08: acceptance makes registration due. Shared 15-minute diagnostic timer.
PSP PENDING	Pinned provider/method processing deadline; payments operations.
PSP COMPLETED	Pinned visibility-check deadline; keep monitoring late outcomes.
PSP FAILED or reconciliation	Immediate owned case; financial handoff where required.
Physical receipt / inspection unfinished	No downstream evidence: D09. Pinned returns policy; example 24 hours from first observation; returns operations.

Payment policies include provider/account, method, region, currency, calendar, timezone, version, and absolute UTC deadlines. No matching policy blocks dispatch; there is no generic processing fallback.

Provider and physical-workflow timers are independent of the UI. Persist their first-observation anchors and deadlines. Repeated snapshots and moves within the evidence-wait group do not reset deadlines.

The fixed 15-minute, 24-hour, and four-hour case-review intervals are illustrative assumptions. Open cases escalate to the owning lead after four hours and receive another review deadline. A handoff requires an accepting owner and case reference.

Test meaning as well as ordering.

Acceptance scenario	Expected behavior
AC-15 · Correct refund, wrong amount	Keep the valid fact; latch reconciliation; open a case.
AC-29 · Rejected item, no-refund closure	RETURN_REJECTED; no false conflict.
AC-69 · Failure after issuance	Accept the newer FAILED; payments case and finance outbox.
AC-76 · Crash after PSP acceptance	Find and bind the existing refund; no new create.
AC-81 · ERP unavailable during correction	Persist the correction and finance handoff; retain reconciliation.

> **112 acceptance scenarios specify expected behavior. They are not executed integration tests.**

The finite model checks 64 state tuples plus 64 reconciliation-flag cases: 128 inputs. It exercises all nine decision rules, checks selected invariants, and compares nine independent input/expected-result examples. There are eight distinct statuses.

These checks do not validate delivery, concurrency, dispatch, provider guarantees, calendars, timers, or ERP integration. Source contracts and operational policies must be verified before production reuse.

Beyond the Scope

Extension	What needs a separate model
Partial refunds / multiple items	Per-item decisions, multiple refund objects, parent aggregation, and concurrent balance limits.
Fees, taxes, shipping deductions, FX	Explicit amount components and currency-specific invariants.
Chargebacks / new refund attempts	Separate objects, authority, transitions, and financial handling.

Per-refund mappings may be reusable where source meanings match. D01–D09 must not be copied as the parent-level rule for a multi-item order.

Review question: which system owns the evidence required for the status we show the customer?

Full specification – see GitHub.

GitHub Repository for the Inside the Spec

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